Reinforcement Learning for Long-Context Reasoning

Hierarchical Agent Architecture with Decomposed Rewards

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The Problem

Mathematical Reasoning Challenge

Large language models struggle with multi-step reasoning [1].
Chain-of-thought helps, but better training methods are needed.

Our Solution

Use RL to incentivize the model towards the right distribution of reasoning and solving behaviors, without a separate value network.

Methods:
Hierarchical decomposition, GRPO [2], decomposed rewards

The Bigger Challenge

Prior work (CoT [1], GRPO [2], hierarchical agents [3]) show promise.

Challenge:
Decoupling rewards between reasoning and solving phases while maintaining signal flow.

Dataset: GSM8K

Overview

- 8.5K grade-school math word problems.
- Linguistically diverse prompts requiring multi-step reasoning.
- Answers include chain-of-thought style reasoning plus a final numeric target.

Data Structure

- **Question:** math word problem.
- **Answer / reasoning:** step-by-step solution logic.
- **Final answer:** numeric label, often marked with `####`.

How We Use It

- Fine-tune on the `train` split.
- Extract exact numeric answers for automated grading and reward computation.

Baseline: Reward-Weighted Supervised Fine-Tuning

Flat Baseline

- Standard causal language model.
- No routing or expert-selection mechanism.
- Custom `RewardWeightedTrainer` for supervised fine-tuning.

Efficiency Choice

- LoRA adapters target attention projections.
- Main targets: `q_proj` and `v_proj`.

Training Objective

1. Format examples as: `Question: [Q] \n Reasoning: [A]`.
2. Assign binary reward from numeric-answer format.
3. Weight sequence cross-entropy by reward.

$$\mathcal{L} = R(x, y)\, \mathcal{L}_{\mathrm{CE}}(x, y)$$

Implementation Note

We identified and fixed an inversion bug in the loss calculation so that successful examples are reinforced rather than suppressed.

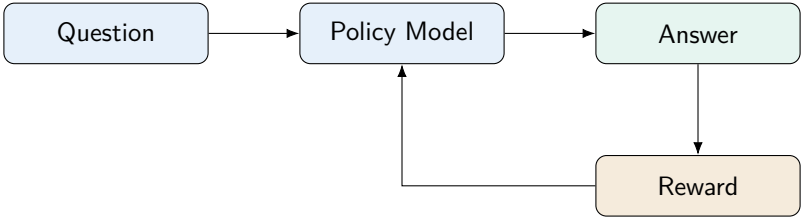
Why Direct RL Over SFT?

Computational Efficiency

- Skip an intermediate SFT stage.
- Optimize directly against task reward.
- Reduce mismatch between training objective and evaluation objective.

Alignment Benefits

- Avoid overfitting to human-curated SFT traces.
- Learn from the reward signal that actually defines success.
- Useful when correct outputs can be automatically checked.



GRPO: Eliminating the Critic Network

Traditional PPO

- Requires a separate critic network.
- Critic estimates per-step value functions.
- Adds memory, compute, and tuning overhead.

GRPO Idea

- Sample a group of responses for the same prompt.
- Compare each response against the group baseline.
- Normalize rewards within the group, avoiding a critic.

$$A_i = \frac{R(r_i) - \mathrm{mean}(R)}{\mathrm{std}(R)}$$

No value model **Lower overhead** **Sequence-level credit**

Router Reward: Outcome-Only vs. Process-Based

Outcome-Only Reward

- $R = 1$ if final answer matches ground truth.
- $R = 0$ if final answer is incorrect.
- Easy to implement and cheap to compute.
- Weakness: gives no feedback on intermediate planning quality.

Process Reward Models

- Score each planning step independently.
- Evaluate clarity, concreteness, and logical order.
- Can detect poor plans before final execution fails.
- Weakness: requires step-level quality metrics.

Core Tradeoff

Outcome rewards are simple but sparse. Process rewards are richer but harder to define without adding noise or reward hacking.

Conditional and Hybrid Router Rewards

Conditional Reward Modeling

- Scores a step based on preceding steps and the final outcome.
- Asks whether the plan would work if executed perfectly.
- Checks for necessary quantities, correct decomposition, and logical order.
- Captures inter-step dependencies but makes credit assignment more complex.

Practical Hybrid Reward

$$R_{\text{hybrid}} = 0.5\, R_{\text{plan}} + 0.5\, R_{\text{outcome}}$$

- Balances structure and correctness.
- Gives smoother learning signal than outcome-only reward.
- Lowest-cost starting point for router reward development.

LLM-as-Judge: Expert Plan Evaluation

How It Works

1. Router generates a structured plan.
2. Send question and plan to Claude with an evaluation prompt.
3. Score clarity, completeness, and feasibility from 0 to 10.
4. Convert to $[0, 1]$ and blend with outcome reward.

Strengths and Limits

- **Strength:** catches subtle plan errors missed by heuristics.
- **Strength:** no reward model training required.
- **Limit:** slower because it requires API calls.
- **Limit:** cost and non-determinism make it risky for high-volume training.

Recommended Use

Train with fast heuristic rewards. Validate with an LLM judge to detect reward inflation, reward deflation, and brittle heuristic behavior.

Challenges & Lessons Learned

1. **Memory Constraints:** Long-sequence decoding (512+ tokens) causes OOM. Hierarchical agent multiplies the problem.
2. **Solution:** Per-rollout backward passes (B=1, G=2) and CodeBatcher for parallel CPU code execution.
3. **RL Training:** Reward design and group size heavily impact convergence. Evaluation cost dominates. Process vs. outcome rewards need careful balance.

Recommended Implementation Path

1. **Stabilize the flat baseline.** Confirm numeric extraction, reward calculation, and loss weighting.
2. **Train a fast router reward.** Start with heuristic plan-quality plus outcome correctness.
3. **Add modular evaluators.** Keep a common `RouterReward` interface so rewards can be swapped cleanly.
4. **Validate with LLM-as-judge.** Use expert scoring for audits, not high-volume inner-loop training.
5. **Compare reward regimes.** Track outcome accuracy, plan quality, reward calibration, and failure modes.

Usage Sketch

```
from src.rewards.router import LLMJudgeRouterReward
evaluator = LLMJudgeRouterReward(model='claude-opus-4-1')
```

References

1 Wei, J., Wang, X., et al. *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models*. NeurIPS 2023. https://arxiv.org/abs/2201.11903

2 DeepSeek Team. *DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning*. 2025. https://arxiv.org/pdf/2501.12948

3 Weng, L. *LLM Powered Autonomous Agents*. Blog Post, 2023. https://lilianweng.github.io/posts/2023-06-23-agent/

4 Zhang et al. *ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search*. NeurIPS 2024. https://arxiv.org/abs/2406.03816

5 *Linking Process to Outcome: Conditional Reward Modeling for LLM Reasoning*. https://arxiv.org/abs/2509.26578